

Emilio Yahir De La Cruz Navarro

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PROFILE

A person who values communication, society, creativity, time, responsibility and initiative. I really enjoy learning and understanding the world. I am open to contributing to the community in any way I am able to and to engaging in environments where curiosity and continuous learning are encouraged.

EDUCATION

Bachelor of Science in Physics, University of Guanajuato, Mexico

Dec 2025

RESEARCH INTERESTS

I am currently interested particularly in astroparticle physics and cosmology with emphasis on neutrino physics and early-universe models. I also have an interest in the conceptual foundations of quantum theory and relativity.

ACADEMIC EXPERIENCE

Research Assistant, Biophotoacoustics Group, University of Guanajuato

Jan 2025 – Dec 2025

Supervised by Dr. Gerardo Gutiérrez-Juárez.

Worked on the replication and adaptation of synthesis protocols for sepia eumelanin and melanin-like nanoparticles derived from dopamine hydrochloride. Investigated nanoparticle size control and dispersion stability in aqueous and biological media. Assisted in troubleshooting UV-Vis spectroscopy instrumentation, including attempts to recover experimental data through serial communication interfaces. Project discontinued before completion due to time and funding constraints.

Student Research Assistant, SNOLAB

Jun 2025 – Oct 2025

Supervised by Dr. Aleksandra Bialek.

Worked with the SNO+ Collaboration on UV-Vis spectroscopy analysis for scintillator compounds and contributed to detector quality assurance (QA) and operational procedures.

Participated in DDA sampling, molecular thin-film still operations, and flushing procedures for the Still Plant, calibration source preparation and deployment (AmBe source and Laser Ball), Deck Clean Room preparation for future detector operations, initiated analysis of low-energy background events and gained experience working in controlled environments using glove bag procedures.

Research Assistant, Quantum Applications Group, University of Guanajuato

Jan 2025 – Jun 2025

Supervised by Dr. Carlos Wiechers Medina.

Contributed to the optimization of a quantum key distribution prototype through analysis of correlated photodetection events.

Collected and analyzed time-interval distributions across multiple detection channels, extracted temporal coincidences, and applied statistical methods to mitigate instrumental limitations and improve data quality.

MITACS Research Intern, Physics Department, McGill University

Jun 2024 – Aug 2024

Supervised by Dr. Thomas Brunner.

Conducted research for the nEXO experiment, a proposed search for neutrinoless double beta decay, focused on improving the optics of an ion trap for barium tagging techniques.

Worked on the reassembly and installation of optical components integrated with an ion detector, initiated characterization of a laser ablation ion source, and performed SIMION simulations of the ex-

perimental setup for comparison with measurements.

PROJECTS

Flat chamber for the detection of ionizing radiation

Aug - Dec 2023

Supervised by Dr. Julián Félix Valdez.

Design, construction, and data analysis of an ionization chamber to detect ionizing radiation by predicting incident particles, especially cosmic radiation, aiming for a better understanding of the quantum effects of muons.

Research Project, Optical Metrology Laboratory

Feb 2023 – Jul 2023

Supervised by Dr. Carlos Wiechers Medina.

Improved measurement techniques for a 633 nm He-Ne laser through dark bead count analysis.

Research Project, COVID-19 Epidemic Modeling with Bayesian Statistics

Jan 2021 – May 2021

Supervised by Dr. Luis Ureña-López.

Studied epidemiological models of herd immunity during the COVID-19 pandemic using Bayesian statistical methods.

TALKS & PRESENTATIONS

SNOLAB Fall Student Talk Symposia

Sudbury, Canada 2025

How QATS help with DDA purification [\[link\]](#)

CASST, Canadian Astro-Particle Physics Summer Student Talk Competition

Sudbury, Canada 2024

Development of a barium tagging technique for the future nEXO detector [\[link\]](#)

TSI Trottier Space Institute Showcase 2024

Montréal, Canada 2024

Development of a barium tagging technique for the future nEXO detector

ElectroWeek 2024

León, México 2024

Presentation of Coulomb's law prototype [\[link\]](#)

CONFERENCES, SCHOOLS & WORKSHOPS ATTENDED

NNN25 [\[link\]](#)

Sudbury, Canada 2025

ENT₁ ENTE [\[link\]](#)

Sudbury, Canada 2025

SNOLAB Underground Science Institute 2025 [\[link\]](#)

Sudbury, Canada 2025

Trottier Space Institute Summer Workshop 2024

Montréal, Canada 2024

6th Summer Particle Astrophysics Workshop by Arthur B. McDonald Institute

Remote 2024

Mexican Cosmology, Particles and String Theory School 2023 [\[link\]](#)

León, México 2023

AWARDS

Concurso Estancias de Verano Experimentales DPyC-SMF 2025

Jan 2025

Selected by the Mexican Physical Society (SMF) for a 16-week funded research internship at SNOLAB within the SNO+ Collaboration.

MITACS Globalink Research Internship 2024

Summer 2024

12-week funded research internship at McGill University.

NASA Space Apps 2023, 3rd Place, Guanajuato Regional Stage.

Nov 2023

NASA Space Apps 2022, 2nd Place, Guadalajara Regional Stage.

Nov 2022

SKILLS

Programming Languages	Python
Languages	Spanish (native), English (fluent), French (basic)
Data Visualisation	pandas, matplotlib
Typesetting Document	L ^A T _E X, Markdown
Additional	UV-Vis spectroscopy, QA procedures, optical setup alignment, glove bag procedures, clean room experience, scientific communication, interdisciplinary collaboration, troubleshooting experimental systems, leadership in student outreach initiatives.

OUTREACH & EXTRACURRICULAR ACTIVITIES

President, GOLF, Organized Group of the Bachelor in Physics	Aug 2024 - Sept 2025
Led a student organization focused on promoting interest in science and technology through outreach activities aimed primarily at students in Guanajuato, Mexico.	
Coordinated the planning and organization of OktoberPhysiFest and BeePy.	
Member, YouthGto	May 2023 – Dec 2024
Participated in initiatives promoting intercultural exchange, leadership development, creativity, and internationalization opportunities for young people in Guanajuato.	